Ayush Tummalapalli

Software Developer | AI/ML Enthusiast | Full-Stack Applications

E-Mail | LinkedIn | GitHub | Portfolio

Education

SRM University AP Aug 2023 – May 2027

B.Tech in Computer Science and Engineering CGPA: 8.23/10

INTI International University, Malaysia Aug 2025 – Dec 2025

Semester Abroad Program CGPA: 3.19/4

Freelancing Experience

Independent Software Developer – Client Project *Vinayaka Brokers*

- Independently designed, developed, and cloud-deployed a brokerage management platform for a business client, now used in daily operations.
- Built features for deal confirmations, brokerage calculations, automated trade contracts, and a registry supporting 200+ company records.
- Implemented automated document generation and PDF/Excel exports to simplify trade documentation, financial reporting, and brokerage workflows.
- Worked directly with the client to gather requirements, demonstrate prototypes, incorporate feedback, pair programmed with AI and delivered a practical business solution.

Projects

Itinera AI – Smart AI Travel Planning Platform *Next.js, React, Generative AI*

- Built and deployed an AI-powered travel planning application that converts natural-language requests into personalized, day-by-day itineraries.
- Developed interactive itinerary customization, budget analysis, weather recommendations, companion preferences, and multi-currency conversion.
- Integrated Gemini API with structured JSON, schema validation, and fallback handling to manage inconsistent AI-generated responses.

IPLytics – AI-Powered IPL Analytics Platform *Python, FastAPI, PostgreSQL*

- Engineered a cricket analytics platform that transforms historical IPL match and ball-by-ball datasets into player, team, and venue insights.
- Developed data-processing and backend workflows using Pandas, SQLAlchemy, and PostgreSQL to support structured data analysis.
- Created interactive Streamlit and Plotly dashboards and integrated Gemini for natural-language exploration of cricket statistics.

Age and Gender Estimation using Dental X-Rays *Python, Machine Learning, Computer Vision*

- Developed a computer vision and machine learning pipeline for estimating age and classifying gender from panoramic dental X-ray images.
- Applied image enhancement, preprocessing, and feature extraction techniques to prepare medical imaging data.
- Evaluated regression and classification approaches to assess predictive performance across age and gender estimation tasks.

Technical Skills

Programming: Python, SQL, JavaScript, Kotlin

Artificial Intelligence: Generative AI, Computer Vision, Machine Learning, Deep Learning, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering

Development & Deployment: Streamlit, React, FastAPI, PostgreSQL, MongoDB, Turso, Render, Vercel

Data & Analytics: Pandas, NumPy, Data Analysis, Data Visualization, Data Preprocessing

Tools: Git, GitHub, VS Code, Google Colab, Google Antigravity, Android Studio, Cisco Packet Tracer, Figma

Certifications

MongoDB Associate Developer Learning Path – MongoDB

- Covered CRUD operations, schema design, document-based databases, and MongoDB Atlas.

CCNA: Introduction to Networks – Cisco Networking Academy

- Covered networking fundamentals, IP addressing, protocols, and Cisco Packet Tracer.